TECHFEST PROTOTYPE • IoT PUBLIC SAFETY

SURAKSHAPATH

Smart Open Manhole Detection & Public Safety Alert System

An IoT-based system that detects open or missing manholes and alerts authorities & citizens instantly.

CATEGORY
IoT Public Safety System

PURPOSE
TechFest Prototype Demo

CENTRE
DigiSkill Khar Road

Keep Roads Safe. Protect Lives. Smart Cities, Safer Cities.

THE PROBLEM

Open Manholes Are a Silent Urban Danger

Invisible Hazard

Open or missing manhole covers on roads and footpaths often go unnoticed until an accident occurs.

Delayed Reporting

Municipal teams typically learn about a missing cover only after a complaint or a mishap is reported.

Life-Threatening Risk

Pedestrians, cyclists, and vehicles are at serious risk of injury, especially during monsoon or at night.

No Early Warning

Existing infrastructure has no automated way to detect a missing cover and alert people in real time.

SurakshaPath solves this with a low-cost ultrasonic sensor and instant LED + buzzer alert — no manhole should ever go unnoticed again.

OBJECTIVE

Key Features of SurakshaPath

Automatic Detection

Detects open or missing manholes using precise ultrasonic distance measurement.

Instant Indication

Green/Red LED indication gives an immediate visual status of the manhole.

Audible Alert

An active buzzer sounds an alert the moment a danger condition is detected.

Low Power

Runs efficiently on a simple 5V supply, suitable for continuous field deployment.

Easy to Install

Built with easily available components — quick to install and maintain.

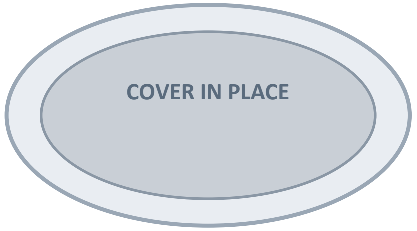
IoT Ready

ESP8266 NodeMCU enables future integration with smart city alert networks.

Two Scenarios: Safe vs. Danger

All electronics are kept outside the manhole in a protected enclosure; only the HC-SR04 sensor sits at the bottom, facing upward.

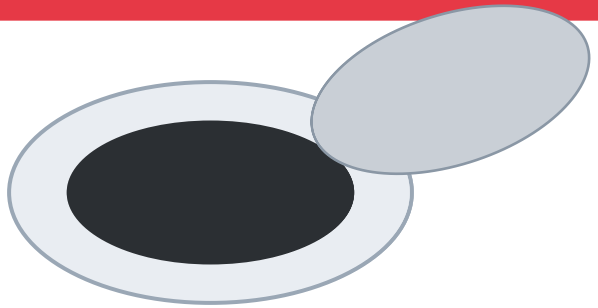
SCENE 1: MANHOLE CLOSED



STATUS: SAFE

- ✓ Distance: 2 – 5 cm
- ✓ Green LED ON
- ✗ Red LED OFF
- ✗ Buzzer OFF

SCENE 2: MANHOLE OPEN



STATUS: DANGER

- ✗ Cover Missing, Distance > 20 cm
- ✗ Red LED ON
- ✗ Green LED OFF
- ✓ Buzzer ON — ALERT

HARDWARE

Components Used

01

ESP8266 NodeMCU

Main microcontroller & Wi-Fi module

02

HC-SR04 Ultrasonic Sensor

Measures distance to detect the cover

03

Red LED

Indicates danger condition

04

Green LED

Indicates safe condition

05

Active Buzzer

Sounds an audible alert

06

Breadboard

Prototyping connections

07

Jumper Wires

Circuit interconnections

08

Power Supply (5V)

Powers the complete circuit

HOW IT WORKS

Working Principle

1. HC-SR04 sensor (bottom of manhole, facing upward) sends an ultrasonic signal.
2. Ultrasonic waves travel upward toward the manhole cover.
3. Waves hit the cover (if present) and echo back to the sensor.
4. ESP8266 NodeMCU processes the reflected signal and computes distance.
5. Based on distance, the system triggers the appropriate LED + buzzer response.

Distance 2–5 cm → SAFE

Cover detected. Green LED ON, Red LED OFF, Buzzer OFF.

Distance > 20 cm → DANGER

Cover missing. Red LED ON, Green LED OFF, Buzzer ON (Alert triggered).

Wiring & Connections

COMPONENT PIN	NODEMCU CONNECTION
HC-SR04 VCC	VIN (5V)
HC-SR04 GND	GND
HC-SR04 TRIG	D5 (GPIO14)
HC-SR04 ECHO	D6 (GPIO12)
Red LED (+)	D2 (GPIO4) through 220Ω
Green LED (+)	D1 (GPIO5) through 220Ω
Buzzer (+)	D3 (GPIO0)
All LED (–)	GND
Buzzer (–)	GND

Signal Flow

HC-SR04 → ESP8266 NodeMCU → Red / Green LED + Buzzer

The NodeMCU reads the echo time from the ultrasonic sensor, calculates distance, and drives the LED/buzzer outputs accordingly.

Note

- All electronic components are kept in a protected enclosure outside the manhole.
- Only the ultrasonic sensor is placed at the bottom of the manhole, facing upward.

WHERE IT FITS

Applications

Smart Cities

Integrates into citywide IoT safety infrastructure.

Municipal Corporations

Enables faster detection and repair scheduling.

Public Safety

Protects pedestrians, cyclists, and vehicles.

Road Maintenance

Flags exact locations needing urgent attention.

Disaster Prevention

Reduces monsoon and night-time accident risk.

THE TEAM BEHIND SURAKSHAPATH

Project Guides & Participants

Project Guides

MK Mr. Karan

MP Ms. Priyanka

Centre: DigiSkill Khar Road

Category: IoT Based Public Safety System

Project Participants

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Thank You

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SurakshaPath — An IoT Based System to Detect Open or Missing Manholes and Alert Authorities & Citizens Instantly

DigiSkill Khar Road | Guides: Mr. Karan, Ms. Priyanka